# The Interrelationships of Unhealthy Food Consumption and Morbidities in Youth: An Umbrella Review

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## BACKGROUND

- Unhealthy food consumption (UFC) is the leading cause of preventable mortality and morbidity.
- Although UFC is linked to increased morbidities in adults, no umbrella review to date examines this relationship within children and adolescents.

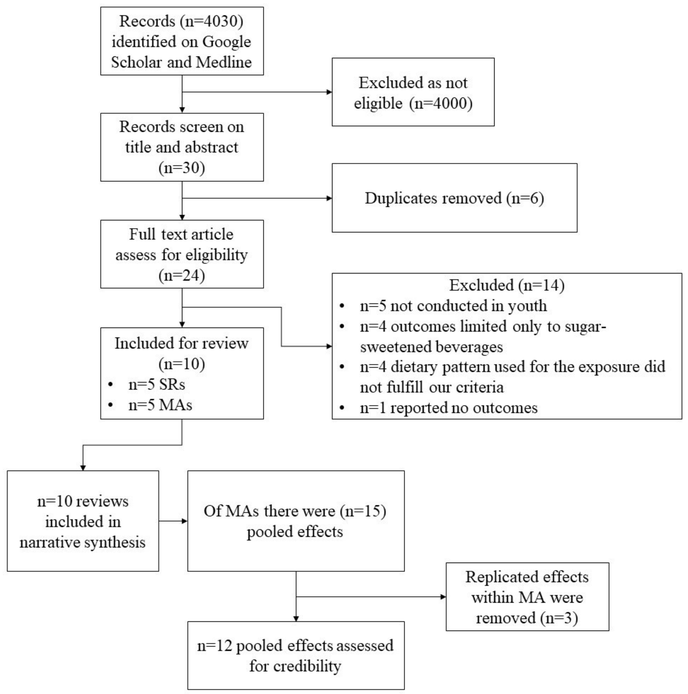
## Aim

- Identify, collect and compare systematic reviews (SR) and meta-analyses (MA) written in English that assess health effects of unhealthy food consumption in youth (<19 yrs).
  - **Exposures:** (**a**) NOVA classification ultra-processed foods, (**b**) WHO unhealthy food list, (**c**) Junk foods, (**d**) Fast foods, (**e**) Western Dietary Pattern (WDP).
  - **Outcomes**: Physical and mental morbidities/risk factors.

## METHODS

- **Exclusions**: Reviews only on sugar-sweetened beverages.
- **Data sources**: Google Scholar and MEDLINE.

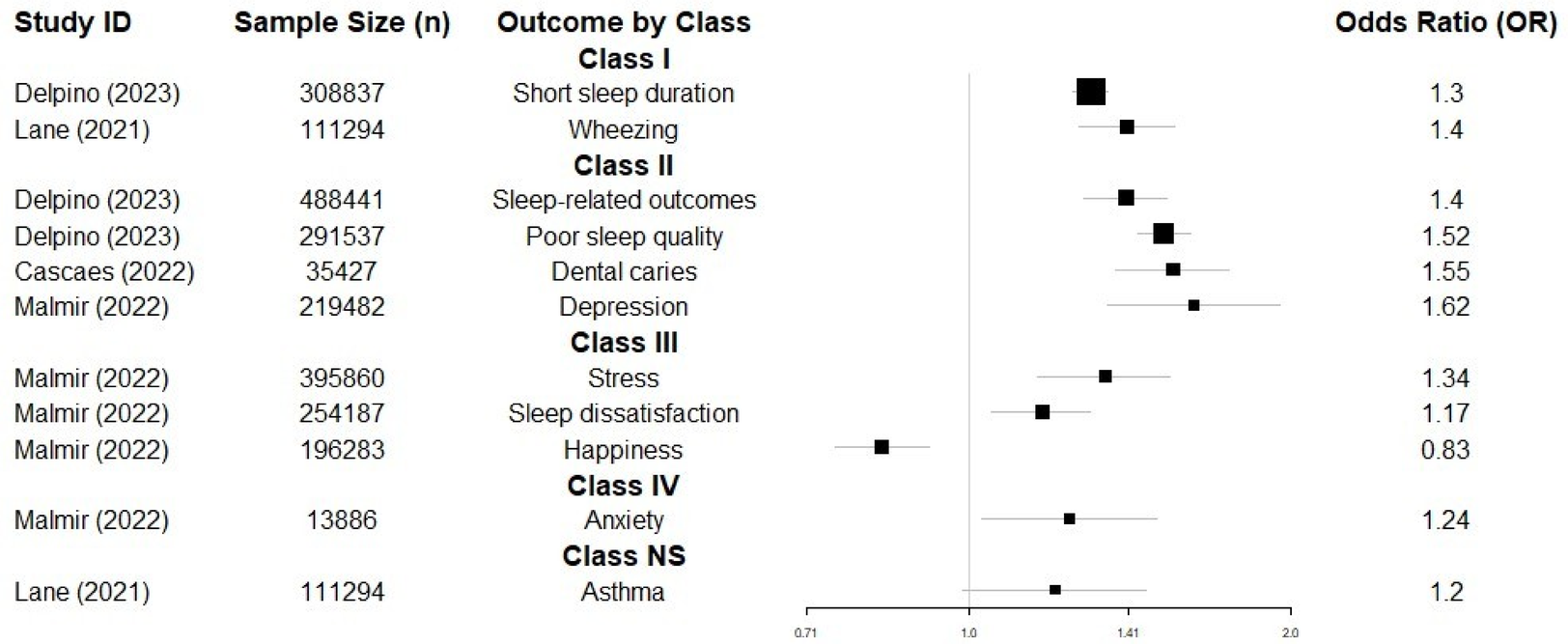
### *Figure 1*: Article Selection Flowchart




*Note.* SR = Systematic Review, MA = Meta-analysis

## RESULTS

### *Figure 2*: Meta-Analyses of Selected Outcomes




*Note.* Criteria for Assessing Credibility Classes – **I:** Convincing (n>1000, P<$10^{-6}$, 95% CI, $I^2$<50%, no small study effects). **II:** Highly Suggestive (n>1000, P<$10^{-6}$, largest study significant). **III:** Suggestive (n>1000, P<$10^{-3}$). **IV:** Weak (P<0.05). **NS:** Non-Significant (P>0.05).

## RESULTS

### Table 1: Methodological Quality of Reviews

| Study ID | Unhealthy Food Classification | Outcome | No. of Studies | N | Quality |
|---|---|---|---|---|---|
| Rousham (2022) | NOVA, WHO, Junk Food, Fast Food | Obesity | 26 | 61903 | High |
| Markey (2023) | NOVA, WHO, Junk Food, Fast Food | CM Risk | 6 | 2469 | High |
| Cascaes (2022) | NOVA | Dental Caries | 42 | 54147 | High |
| De Amicis (2022) | NOVA | Obesity | 10 | 24281 | Moderate |
| Bujtor (2021) | WDP, Fast Food | Inflam. Bio. | 3 | 52399 | Moderate |
| Delpino (2023) | NOVA | Sleep | 8 | 488441 | Moderate |
| Costa (2018) | NOVA | Obesity | 11 | 15208 | Moderate |
| Lane (2021) | NOVA | NCD | 9 | 144461 | Low |
| Rocha (2017) | WDP, Fast Food | CM Risk | 9 | 11737 | Low |
| Malmir (2022) | Junk Food | Psy. Distress | 17 | 263099 | Critically Low |

*Note*. **Unhealthy Food Classes**: NOVA=ultra-processed, WHO=World Health Org. unhealthy food list, WDP=Western Dietary Pattern. **Outcomes**: Obesity, Inflam. Bio.=Inflammatory Biomarkers, CM Risk=Cardiometabolic Risk, Sleep, Dental Caries, NCD=Non-Communicable Diseases, Psy Distress=Psychological Distress. **Methodological quality** was assessed using AMSTAR 2.

## CONCLUSIONS

- Convincing evidence links UFC to short sleep duration and wheezing.
- High methodological quality reviews link UFC to obesity, cardiometabolic risk, and dental caries (highly suggestive).
- Critically low methodological quality suggestive/weak evidence associates UFC with psychological distress.
- Further research is needed to assess relationships between UFC and psychological outcomes in youth.

References

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